Host Utilities

TATA

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

ORGANIZER

ISGF
India Smart Grid Forum

India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Supporting Ministries

सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA

सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

Session : 13th March 2026

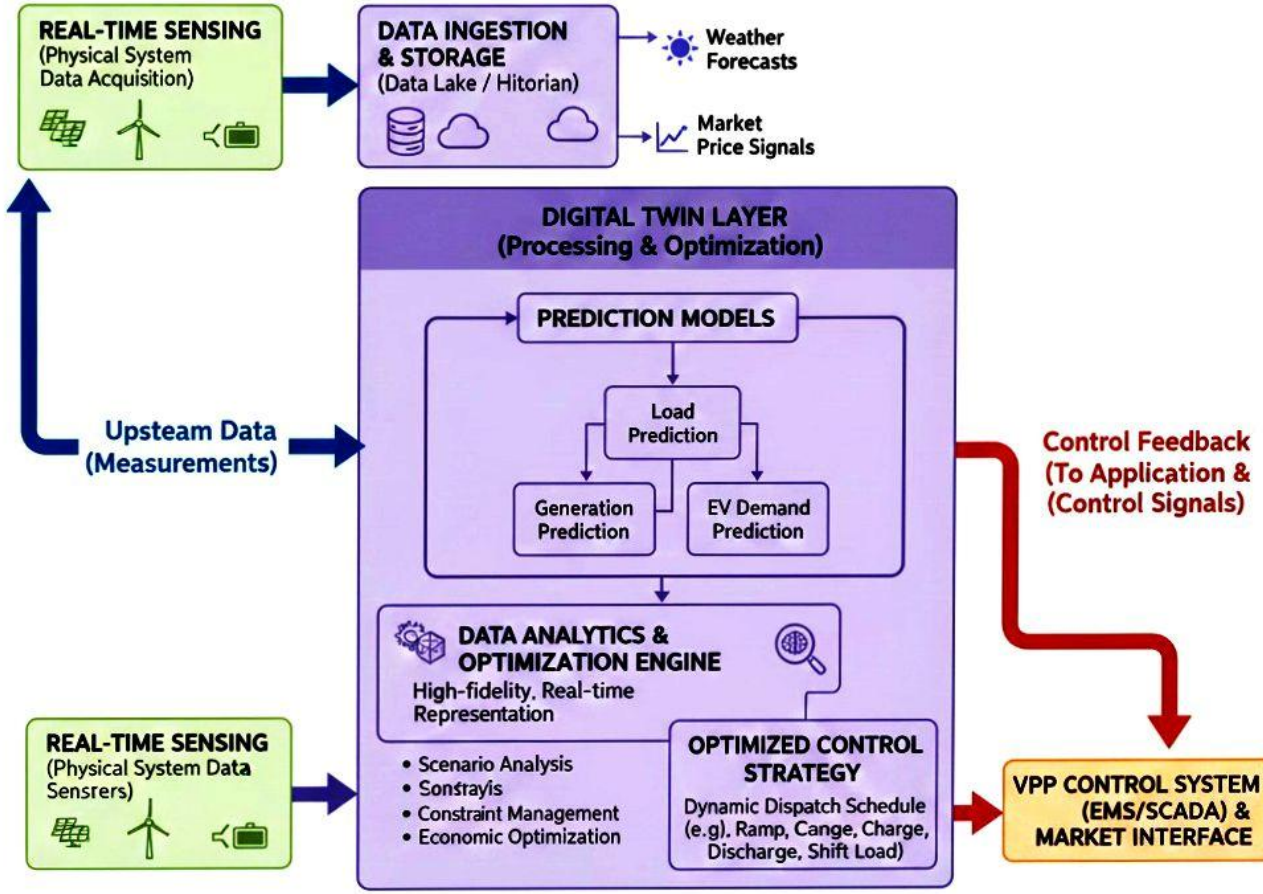
Digital Twin-Enabled Virtual Power Plants for Optimal Renewable Energy Integration and Grid Flexibility in India

Presented By

Dr. T. Kesavan, Assistant Professor, Department of Electrical and Electronic Engineering, Easwari Engineering College, Chennai

INTRODUCTION

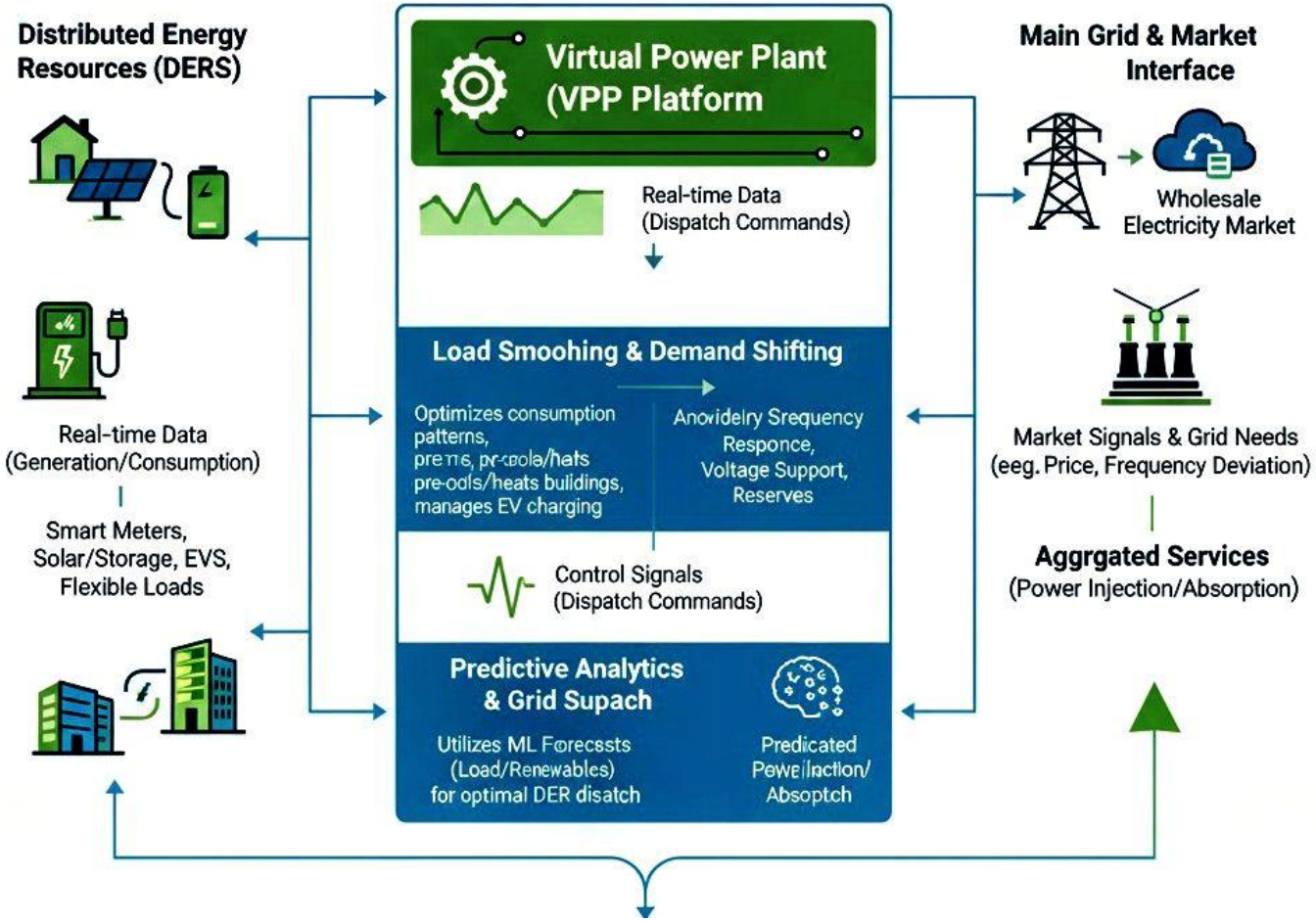
- National Targets: India aims for 500 GW of renewable capacity by 2030 and net-zero by 2070.
- The Problem: Variability of RES causes voltage instability and reverse power flows in conventional grids.
- The Digital Twin Solution: A virtual version of the grid that uses IoT and SCADA data to forecast and optimize assets.
- Expected Outcome: Scalable solutions for resilient power systems.



Functional diagram for digital twin based VPP

CONTEXT

- Energy Shift: Transitioning from centralized fossil fuels to decentralized "swarm intelligence".
- VPP Utility: Aggregates DERs like solar, wind, and EVs into a single dispatchable unit.
- Technical vs. Commercial: Differentiates between CVPP (market participation) and TVPP (local grid support).
- Indian Regulatory Catalyst: Driven by new 5-minute scheduling and stricter Deviation Settlement Mechanism (DSM) norms



Grid Flexibility Enhancement in Virtual Power Plant

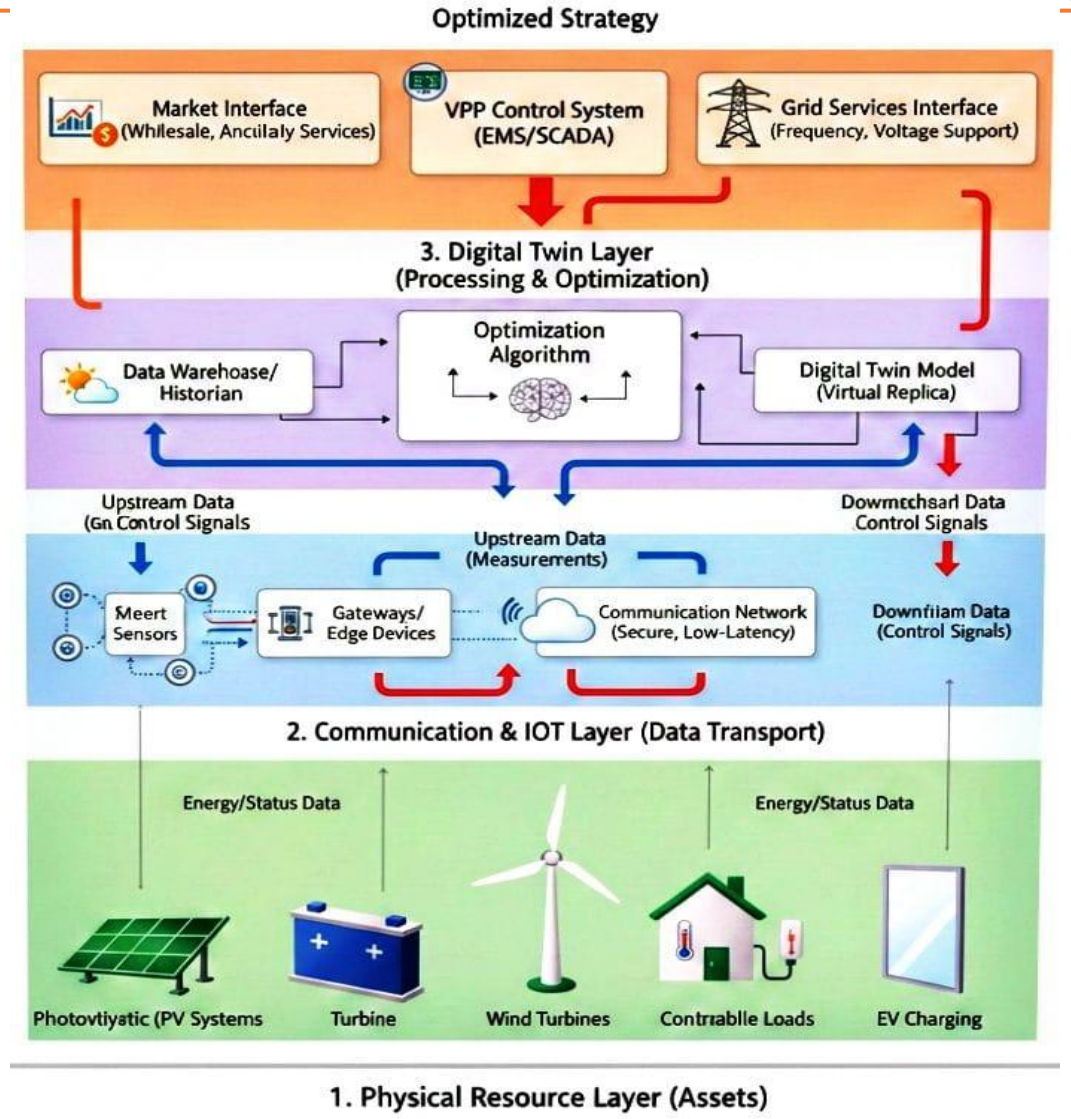
RELEVANCE

- Hosting Capacity: DT-VPP increases solar injection by up to 20% over traditional grids.
- Voltage Stability: Reduces maximum voltage deviation from 6% to 2.5%.
- Loss Mitigation: Technical line losses decreased by 15.5% daily.
- Forecasting Accuracy: Machine learning achieves a Mean Absolute Percentage Error (MAPE) of 8% vs. 15-20% in traditional models.

Metric	Traditional Grid	DT-Enabled VPP	Improvement
Max PV Penetration	2.5 MW	3.0 MW	+20%
Max Voltage (p.u.)	1.08 (Violation)	1.045 (Safe)	Compliant
Line Losses (Daily)	450 kWh	380 kWh	-15.5%

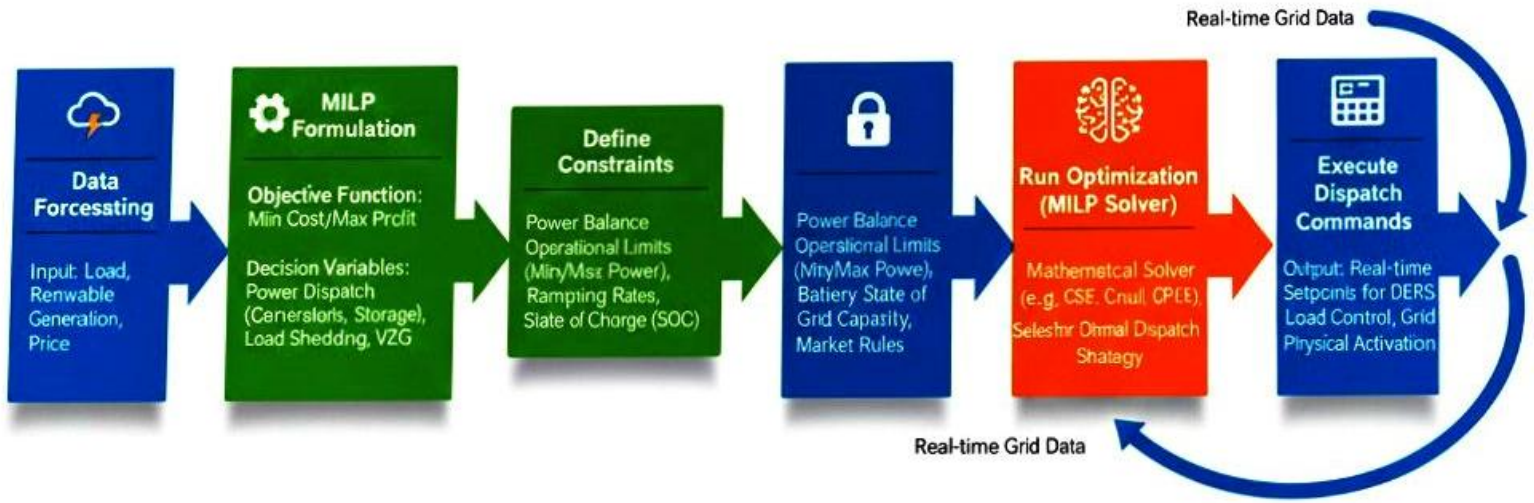
ARCHITECTURE

- Physical Layer: Includes smart inverters, BESS, and 2/3-wheeler EVs acting as "batteries on wheels".
- Communication Layer: Hybrid 5G/4G network using MQTT and IEC 61850 protocols for ultra-low latency.
- Digital Twin Layer: Synchronized virtual models execute CNN and LSTM algorithms for predictive analysis.
- Application Layer: Facilitates V2G services and peak load adjustment via OpenADR.



Proposed Digital Twin based Virtual power plant

OPTIMIZATION

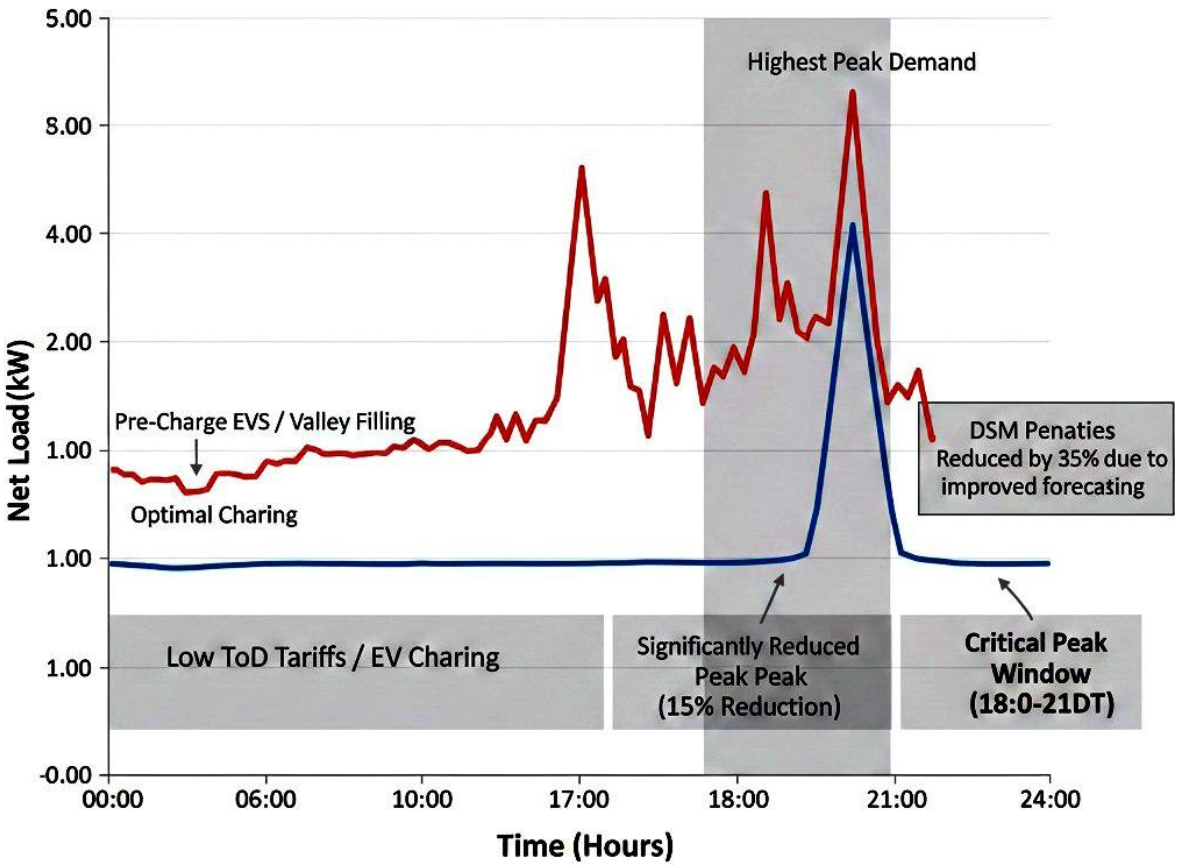


MILP based optimization workflow diagram

- Predictive Analysis: Uses satellite imagery and time-series data to anticipate weather impacts.
- MILP Workflow: Mixed-Integer Linear Programming optimizes revenue while respecting physical grid constraints.
- Asset Health: Detects fault precursors in transformers up to 30 hours before failure.
- Coordinated Operation: Optimizing solar, wind, and storage together yields better outcomes than individual optimization.

USE CASE / CASE STUDY

- Tamil Nadu: Reduced curtailment in hybrid solar-wind feeders using BESS.
- Gujarat: VPP-enabled rooftop solar aggregation for effective Time-of-Day tariff participation.
- Delhi Microgrid: Lowered peak demand by 15% and reduced DSM penalties by 35% via Proactive Control.
- V2G Stability: EV fleets monitored by DT provided rapid primary frequency response during grid dips



Power Dispatch curve on Delhi Urban Microgrid (24 Hours).

Host Utilities

TATA

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

ORGANIZER
ISGF
India Smart Grid Forum

India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

Supporting Ministries

सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA

सत्यमेव जयते
MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)